Investigating Zero-shot Topic Labeling of Scientific Papers Using LLMs

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Abstract: In this paper, we focus on the problem of adding content labels of a given vocabulary to scientific publications using LLMs. After a short overview of the current state of the work, we present a first implementation of a zero-shot classification pipeline. This implementation is already realized with a focus on extendibility and customizability, so that it can easily be used for different data sets and use cases in the future. We select a subset of the DBLP Discovery Dataset and execute our pipeline on it. In the end, we discuss the results, suggest a comparison with a second data set, the STTCL journal from the humanities, and present its challenges. Both of the mentioned data sets comply with the FAIR data principles. Finally, we consider our plans for the next steps.

Keywords: LLM, automatic annotation, natural language processing, text analytics, annotation with a controlled vocabulary, AI-enabled methods, AI-readiness for research data, hierarchical multi-label classification, text classification, FAIR data principles

1 Introduction

Nowadays, the number of released scientific publications increases steadily. This development roots in different reasons, for example the rapidly growing number of researchers in most parts of the world. On the one hand, this advance is highly valuable. On the other hand, it makes it increasingly difficult to gather an overview of the research that has already been conducted. As a consequence, more and more work time is spent with creating such an overview instead of gathering new results. Therefore, the urgent need for a simplification of this process is very obvious. One such simplification would be the annotation of documents with topics from a controlled vocabulary. This would simplify the search for publications covering a specific topic. A controlled vocabulary offers better comparability and quality, since the opposite approach does not involve manual reviews.

The problem of automatically annotating documents with a controlled vocabulary is not new but has already been studied with multiple approaches (see for example [Fe97], [PSI06]). We try a modern approach, however, and discover a surprising result.

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Recent progress in the field of natural language processing provided us with a new technology known as *large language models (LLM)*. These models learn statistical relationships in their training corpus and are usually trained on large amounts of text data. A key feature of these models is that they can parse free-text prompts and even give responses to those. Furthermore, there can be contextual information provided in the query itself. For instance one could ask the model to summarize a paragraph of text that is provided in the query. It is important to note that these models are not simply trained on summarizing text but rather have a more universal set of skills.

In this paper we discuss how we can leverage a LLM in order to automatically annotate documents with topics from a controlled vocabulary. Therefore, we want to classify scientific publications from the computer science domain into topics from a controlled vocabulary. First, we derive a dataset of already labeled documents from the *DBLP Discovery Dataset (D3)*[3] created by Wahle et al. [Wa22], which provides automatically generated labels from the *Computer Science Ontology (CSO)*[4], a project created by Salatino et al. [Sa18]. We restrict our attention to only a small amount of the labels from the CSO, which form our controlled vocabulary of topics. We create a classification task and run two separate experiments of it, one using the titles as document representation, the other one using the abstract. Our workflow is to construct a set of prompts to provide the model with the possible topics and then ask the LLM which of those fits the given document best. In our experiments, we used the Llama 3 model, a state-of-the-art LLM published by Meta Platforms. Finally, we run an evaluation comparing the obtained annotation from the LLM approach with the already provided labels in the D3 dataset.

Our results show that the rate of publications that are annotated with a correct topic (according to our evaluation data) is the same for both titles and abstracts as document representation. Surprisingly, we observed that we obtain a significantly higher hallucination rate when abstracts are used instead of titles, which means that the LLM returns something that can not be matched to a topic from our controlled vocabulary. Both of these results were quite counter-intuitive to us because we initially assumed that the abstracts would yield richer semantic information than the titles and thus produce better results.

The paper is structured as follows. In Section 2, we discuss the current state of research regarding automatic annotation with controlled vocabularies. Section 3 discusses how we obtained the dataset used in this evaluation. In Section 4, we briefly describe how we constructed the prompts for the LLM and mapped the results on the controlled vocabulary. The technical details of our implementation and the LLM used in this project can be found in Section 5. An evaluation of the results obtained by the annotation with the LLM is discussed in Section 6. Finally, Section 7 concludes the paper and discusses potential future work.

3 https://zenodo.org/records/7071698
4 https://cso.kmi.open.ac.uk/about

2 Related work

There has been a long interest in adding topic labels to texts, also known as text classification or topic labeling. Classic approaches for topic labeling used, for example, support vector machines (SVM) or naive Bayes classifiers. For a more detailed overview, we refer to e. g. [Ko19]. However, since the recent rise of Deep Learning, the focus has shifted to such approaches, e. g. transformers like BERT [De19] or large language models (LLMs), cf. [Li22], [Ta24b] or [ZJN21].

Obviously, a "non-hierarchical single-label classification" approach such as the one presented below can also be seen as a special case of *hierarchical multi-label classification* (where in general the labels represent the leaf nodes of a multi-tier hierarchy tree and documents are always labeled with the leaf nodes and also the nodes of the path): The used labels from the CSO are nothing less than one layer of the label hierarchy of the CSO and since there is only one such layer within our labels, a path would simply contain not more than one node: A leaf node. In [SC22], Sadat et al. used the ACM CCS tree as labels when trying a similar idea exclusively on computer science literature. Hierarchical multi-label classification has been implemented even at an industrial scale, see [Ta24a]. Martorana et al. tried a variant by adding labels from a controlled Linked Data vocabulary to column headers using LLMs [Ma24]. For our early approach, a single layer of labels will be sufficient, but hierarchical levels of labels will be a relevant option for future work.

It should be noted that the term "topic" is used for multiple concepts: The well-known discipline of topic modeling uses it for sets of terms which are automatically generated during the execution. Our method however uses existing and established vocabularies, which has the advantage that we do not introduce any new incompatibilities with the existing vocabulary. It can be noted that our use case can also be distinguished from Named Entity Recognition (NER), since the topics might not be mentioned explicitly in the title or abstract.

3 The dataset

In order to test an automatic annotation approach based on LLMs, we needed some data that is already labeled in some form. For obtaining such a dataset, we used the aforementioned dataset D3. It contains all papers and authors extracted from the *dblp computer science bibliography*,[5] which is operated by NFDI4DS and NFDIxCS member Schloss Dagstuhl LZI,[6] and enriched these records with further metadata: For example, abstracts have been added to publication records. Since its original release, there have been two new versions of the dataset, where the latest version (2.1) incorporates labels from the aforementioned CSO. More information about the new versions of D3 can be found on the project GitHub page.[7]

5 https://dblp.org

6 https://blog.dblp.org/2021/07/02/schloss-dagstuhl-becomes-part-of-the-national-research-data-infrastructure-for-data-science-and-artificial-intelligence/

7 https://github.com/jpwahle/lrec22-d3-dataset

The CSO is a large scale hierarchical ontology of research areas in computer science, but also in other domains as well: Therefore, the ontology contains secondary root nodes like “linguistics” or “geometry” in addition to the self-explanatory root node “computer science”. At the time of writing, “computer science” has 19 subordinate topics, which are:

- artificial intelligence
- computer aided design
- computer hardware
- computer imaging and vision
- computer networks
- computer programming
- computer security
- computer systems
- data mining
- human computer interaction
- information retrieval
- information technology
- internet
- operating systems
- robotics
- software
- software engineering
- theoretical computer science
- bioinformatics

These 19 topics form the controlled vocabulary for the annotation process in our approach.

The documents in the D3 dataset have been labeled with topics using the CSO classifier.[8] This classifier has two separate modules, the syntactic and the semantic one, and outputs four sets of labels. The first two sets correspond to the outputs of each module, the third is their union and the fourth includes the relevant super-areas in the ontology. Thus, when building the union over all four lists, we should have at least one of the high level topics presented above for every classified document.

By using this assumption, we created our test dataset from the D3 dataset in the following way: We considered all publications in the publications part of the dataset. For every publication, we computed the union of all four resulting lists of the CSO classification. Then, we discarded all elements of this list besides the 19 high level topics we discussed earlier. If the list obtained this way has no elements, the publication is discarded. The minimal number of topics is thus one. It should be noted that a publication can have more than one topic associated with itself: The maximal number of topics observed in our dataset is twelve. On average, a publication has approximately 4.5 topics. Then, we filtered out all publications that did not have a title or an abstract. From the resulting publications, we selected 2500 at random, which were used to form our dataset.

The extracted dataset is saved in a JSON file. This file contains a list of dictionaries, where each dictionary represents one of the 2500 randomly selected publications with a small set of metadata, which can be seen below. For our purposes, we saved the `corpusid`, which is the identifier of the publication in the D3 dataset under the `D3 ID` key and saved the title,

8 `https://github.com/angelosalatino/cso-classifier`

```
{
  "D3 ID": 18298,
  "title": "Harnessing the power of two crossmatches",
  "abstract": "Kidney exchanges allow incompatible donor-patient ..."
  "subjects": [
    "computer hardware",
    "computer networks",
    "computer systems",
    "information retrieval",
    "information technology",
    "internet",
    "theoretical computer science"
  ]
}
```


Fig. 1: First publication record of the dataset extracted from D3 (abstract shortened)

the abstract and the extracted topics in the `title`, `abstract` and `subjects` keys respectively. An example of such a publication entry is given in Figure 1. Even though the abstract might seem like a misclassified item from medicine at first glance, it is indeed a computer science publication.

4 Methodology

Now that we have an understanding of how our data was obtained, we will discuss the methodology for the annotation process using an LLM. In order to annotate our documents, we need to create a set of prompts for each document that is processed by the LLM, such that we obtain a "parsable" answer from which we can extract the annotated data. For the setting presented in this paper, we focused on using the titles and abstracts in separate experiments as document representation data and wanted to obtain one of the topics from the controlled vocabulary constructed in Section 3 that fits to the document in each experiment. We decided to use three prompts for every document: The first two are used to provide the model with our controlled vocabulary, the 19 subordinate topics of "computer science" from the CSO, and the last one asks to annotate the current document. The concrete prompts (in the version for the experiment using the title as document representation) are presented in Figure 2, where `{document representation}` represents only a placeholder for this figure and is substituted with the title of the current document in the implementation. The prompts for the experiment based on the abstracts are analog in structure: We only replaced the word "title" with the word "abstract" in the static part of the prompt and provided the LLM with the abstract of the document instead of the title at the `{document representation}` placeholder.

These three prompts were evaluated for every document in our dataset. As it can be seen in the last query, we informed the LLM that we only wanted one topic from our controlled

We want to create a list of topics in the following. We call this list targets_list.

Here are some topics that should be added to the targets_list: artificial intelligence, computer aided design, computer hardware, computer imaging and vision, computer networks, computer programming, computer security, computer systems, data mining, human computer interaction, information retrieval, information technology, internet, operating systems, robotics, software, software engineering, theoretical computer science, bioinformatics. Please use the exact spelling that I provide to you.

We now want to annotate a title with the topics provided in the targets_list.
Given the following title: `{document representation}`
Please assign 1 suitable topic from the targets_list to the title.
This topic should be contained in the targets_list we created earlier and use the exact spelling of the topic in the targets_list.
Please respond only with the 1 topic without any further text.

Fig. 2: Prompts used for this work where the term `{document representation}` represents a placeholder which is replaced with the actual title in every request

vocabulary as an answer. Then, we took the result of the third query and tried to match it with our controlled vocabulary in the following way: We split the result string at the symbol "," and treated each of the elements obtained that way as a candidate string that should be matched to our controlled vocabulary. Then, we transformed each of these candidate strings to lower case and removed all whitespace at the beginning and the end. It should be noted that the string splitting and the removal of the whitespaces was done because the code used for this evaluation is also capable of asking the LLM for more than one topic and in that case, it expects a comma-separated list as result, whereas the normalization in terms of converting the string to lower case was introduced because we observed that some models used capital letters in the response. We matched a candidate string to an element in the controlled vocabulary if the strings were equal.

5 Implementation

The code together with the instructions on how to perform our analysis is provided on GitHub.[9] To foster its further usage, we have put efforts in making the tool *FAIR* (findable, accessible, interoperable and reusable) [Wi16] by using an open license, hosting the code on GitHub and also the usage of open tools and packages. The code is written in Python, using the GPT4All package[10] [An23] for all LLM related aspects.

9 `https://github.com/paNeises/paper-2025-bigds`
10 `https://pypi.org/project/gpt4all/`

For better reproducibility, self-hosted solutions were preferred instead of cloud-based services like ChatGPT or Claude. From these self-hosted software suites, we preferred a tool that not only could provide access to multiple LLMs but also could be accessed via existing SDKs, bindings or program libraries, and was likewise mature and comparably well-documented. Taking these requirements into account, GPT4All fulfilled our requirements. We used it via its Python bindings (version 2.8.2).[11]

In order to annotate the dataset, our implementation constructed the three prompts described in Section 4 for every document in the dataset. The prompts were then evaluated with an LLM using a separate chat session for every document. GPT4All allows to use multiple LLMs out of the box. It has to be noted that there exists the option to include further user-embedded LLMs, which is another interesting question for future work. Therefore, it was not possible to test every LLM that could run with GPT4All. We performed tests with predefined LLMs and eventually chose Llama for this first implementation, since it showed decent results. We used the Meta Llama 3 model (`Meta-Llama-3-8B-Instruct.Q4_0.gguf`), which is already preconfigured as one option with the GPT4All Python bindings.

For each document, the annotated keywords obtained by the LLM were then added to the dictionary representing its metadata as a new field and the final list containing all documents was stored. Finally, we ran an evaluation on the previously saved annotated set of documents, whose results are covered in Section 6.

6 Evaluation

The annotation as well as the evaluation were executed on a workstation equipped with an AMD Ryzen 9 7900X with twelve cores and 24 threads, 128 GB of RAM, a single Nvidia RTX 4090 and a 2 TB M.2 PCIe SSD.

In order to evaluate the annotated data obtained by the LLM, we differentiated the following three cases when observing the annotation result:

- A topic from our provided topic list was successfully extracted from the LLM response and this topic is in the subjects-list in the document metadata. We call this case *success*.
- A topic from our provided topic list was successfully extracted from the LLM response but this topic is not in the subjects-list in the document metadata. We call this case *misclassified*.
- It was not possible to extract a topic that matches one of the entries in our topic list from the LLM response. We call this case *hallucination*, which might differ a bit from its common usage.

11 Please beware that the semantic versioning of GPT4All differs from the semantic versioning of the Python bindings published on PyPi. We refer to the version of the Python bindings, cf. `https://pypi.org/project/gpt4all/#history`, which have not yet reached version 3 and beyond.

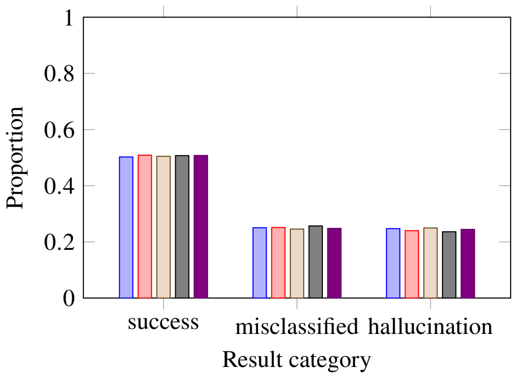

Fig. 3: Analysis of the LLM annotation on the dataset extracted from D3 based on titles (multiple runs indicated by color; each bar represents the proportion of a result category from a single run)

It should be noted that after the matching step, the returned list of keywords that were annotated by the LLM contains one element for the first two cases, whereas the last case always implies that the list is empty. As previously mentioned, we conducted two separate experiments, the first of which using the document title as the document representation and the second one using its abstract. The workflow of these experiments was already discussed in Section 4 and Section 5. We gathered the results from the annotation process using the LLM and compared the results with the topics in the subject-lists in the document metadata extracted from D3 in the process of creating the dataset as discussed in Section 3. In every experiment we performed five separate runs in order to see if there is a variation between the results of these runs.

The results of the first experiment (based on titles) are shown in Figure 3. As we can see, the results remained fairly consistent between the runs in terms of numbers. We observed the success case in around 50 % of the documents labeled by the LLM across all runs. The misclassified case, on the other, hand occurs in 25 % of the documents. For 25 % of the documents, the annotation by the LLM ends up in the hallucination case. The running time of the annotation process of all 2500 documents also remained consistent across all runs and took us around 2 hours and 52 minutes for each run.

The results of the second experiment based on abstracts are shown in Figure 4. Similarly to the results based on the titles, we also observe that the results stayed consistent between the different runs in terms of numbers. Also, the rate of the success case did not change and stayed at around 50 %. On the other hand, we did not observe that many misclassified cases in the second experiment, as their rate is only about 17 %. This leaves around 33 % of the documents, for which we ended in the hallucination case. The running time again stayed consistent between all runs and took us about 3 hours and 30 minutes for each run.

These results were quite surprising to us. We did not anticipate the following three aspects:

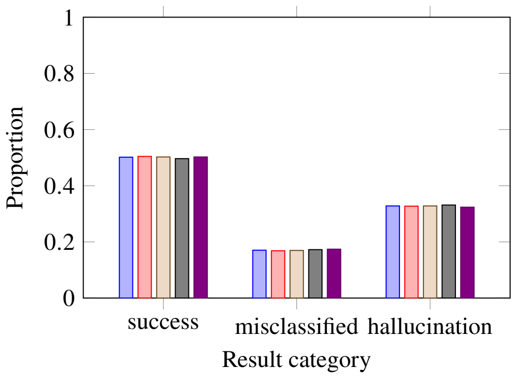

Fig. 4: Analysis of the LLM annotation on the dataset extracted from D3 based on abstracts (multiple runs indicated by color; each bar represents the proportion of a result category from a single run)

On the one hand, we did not expect the decrease of the success rate, since the abstracts usually provide much more details and context than the title, which could help to classify the documents correctly. On the other hand, the increase in the hallucination rate itself was unexpected. Our assumption was that it would at least stay constant or even go down, since the requirement, that the given vocabulary had to be used, did not change. At last, we did not assume that the number of results would shift so significantly from results of the case misclassified to results of the case hallucination, as we anticipated a growth of the success rate.

We further inspected some of the responses that led to a hallucination case by hand using a separate annotation run based on titles. For example, we observed in one case that the LLM returns "'software engineering'" (using the inner quotes; outer quotation marks due to the quotation), which prevents our matching step from finding the correct topic since it does not remove quotes. Further, we observed multiple responses from the LLM that were "computer vision". However, this topic does not exist in our controlled vocabulary but "computer imaging and vision" does. In these cases, it would be possible to resolve these cases with a better matching step. Nevertheless, by far the most answers, that we observed and that ended up in the hallucination case, were "computer science".

From a statistical point of view, a random guessing algorithm would provide about 24 % success rate, since we have 19 topics and each document is annotated with 4.5 topics on average. Compared to such an algorithm, this rate is significantly better, even though it is not as high as we expected it to be. This leaves room for discussion and considerations to improve this experimental workflow. Possible reasons might be that the subject areas of scientific topics are too small and insufficiently covered in the training data sets, so that LLMs do not reach the results achieved in other areas.

7 Discussion and future work

As shown in Section 6, we provided a zero-shot annotation approach that significantly outperforms a random guessing algorithm on the discussed dataset. We present it as an early-stage approach together with the lessons learned and are looking for scientific feedback for further development. The results indicate room for improvement of the presented approach. In the following, we discuss some ideas for future work.

Our results show that the rate of publications annotated with a correct topic stays the same for both titles and abstracts as document representations. An explanation might be that more input also leads to more possibilities that could "confuse" the model, which is an interesting question for future research. Also, we observed two limiting factors for our approach: The *suggested maximal prompt length* (maximum length of a prompt that the LLM was trained for) and its *context length* (maximum previous input/output that the LLM considers for answers). However, the documentation lacks information about them. We only observed these limits in the debug messages, rises interesting questions for the future. Additionally, we only focused on a small set of topics extracted from a large scale ontology. In the future, we want to explore the scalability in terms of the number of topics.

Another interesting question is if the model can correct itself in the case of a hallucination. This case is easily detectable because we can apply our matching step and consider an empty result as hallucination, since we did not obtain a topic from our controlled vocabulary. The idea is for such cases to supply an additional prompt to the LLM that points to the erroneous output and asks the LLM to correct it to a topic from the provided vocabulary.

Our intention was to create this framework as customizable as (sensibly) possible, so that it could be used by other people in different domains as well. We have also gained the impression that LLMs for topic labeling are not frequently used in the humanities so far. Therefore, we wanted to test it also with data from the humanities, but it turned out that such (coherent) digital text or metadata sets together with suitable topic vocabularies are quite rarely available and are even more unlikely to have an open license. We found the journal "Studies in 20th & 21st Century Literature (STTCL)",[12] which uses an open access license. Its articles are already labeled with topics from the Digital Commons Network™ (DCN),[13] which seemed suitable for this approach. We took the union of all DCN topics annotated to STTCL documents as the set of which the LLM had to choose a valid topic. First tests with the titles of all items of the journal (the issues include book reviews, editor's notes, etc.) showed only a success rate of about 10 %, which might indicate that this data sets requires some preprocessing before it is usable. Therefore, we plan to tackle this approach again with a further processed version of the STTCL data set in the future. As a first result from these investigations, it can be concluded that (open) data sets in the humanities are often in an earlier stage than data sets in more computer-related disciplines and might require more manual preprocessing.

12 `https://newprairiepress.org/sttcl/`
13 `https://network.bepress.com/`

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